

Final Capstone Project Report

Mutual Fund Analytics & Quantitative Risk Platform (Days 1–6)

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Branding Theme: Bluestock Navy & Cyan

Database System: SQLite 3

BI Integration: Power BI Desktop (Deep Carbon Dark Theme)

1. Executive Summary

This report presents the complete end-to-end design, implementation, and analysis of the Mutual Fund Analytics & Quantitative Risk Platform. The project establishes a robust data architecture for mutual fund managers, investment analysts, and retail client advisors to monitor industry market share, run deep risk-adjusted performance comparisons, track retail investor cohorts, and evaluate portfolio tail-risk exposures.

Key Objectives Accomplished:

- Production ETL Pipeline: Engineered a Python-based pipeline that cleans raw NAV histories, investor transaction logs, and performance metrics, fetching live AMFI NAV rates and Yahoo Finance index rates (^CNX100, ^NSEI).
- Relational Database Design: Implemented an SQLite Star Schema with dimension constraints, automatic indexes, and analytical views to optimize query speeds.
- Exploratory Data Analysis (EDA): Rendered 17 interactive (Plotly) and statistical (Seaborn/Matplotlib) plots, exporting them to reports for multi-channel distribution.
- Performance & Risk Scorecards: Programmed quantitative engines to compute CAGRs, Sharpe/Sortino ratios, CAPM regression alphas/betas, Maximum Drawdowns, Tracking Errors, Historical Value at Risk (VaR), Conditional VaR (CVaR), and Herfindahl-Hirschman Index (HHI) sector concentrations.
- Power BI Visual Layouts: Designed a 4-page responsive executive dashboard specification under a custom Bluestock Navy branding system.

2. Data Sources & Integration Strategy

The platform ingests five primary datasets to establish a unified single source of truth:

```
graph TD
  A[AMFI NAV Files] -->|Clean & Ingest| D[(SQLite DB: mutual_funds.db)]
  B[Investor Transactions] -->|Validate & Load| D
  C[Yahoo Finance API] -->|Index Close Prices| D
  E[Scheme Performance Records] -->|Quarterly Ratios| D
```

```
D --> F[EDA Module: 17 Charts]
D --> G[Performance Module: Scorecard]
D --> H[Risk Module: VaR, HHI, Cohorts]
```

Data Dictionary Summary

1. `nav_history.csv` (AMFI / Daily NAV): Historical Net Asset Values (NAV) for 10 distinct Indian mutual fund schemes spanning 2022-01-03 to 2024-12-31 (7,798 total daily observations). Key columns: `scheme_code`, `nav_date`, `nav`, `repurchase_price`, `sale_price`.
2. `investor_transactions.csv` (Retail Inflow Ledger): Granular log of 1,985 transactions across 200 distinct retail investors. Columns: `txn_id`, `investor_id`, `scheme_code`, `transaction_type` (SIP, Lumpsum, Redemption, Switch-In, Switch-Out), `amount_inr`, `units`, `nav_at_txn`, `transaction_date`.
3. `scheme_performance.csv` (Quarterly Snapshot): Historical returns (1M, 3M, 6M, 1Y, 3Y, 5Y), AUM (in Crores), and expense ratios.
4. Yahoo Finance Index Feeds (`nifty50.csv`, `nifty100.csv`): Historical daily close prices for Nifty 50 (^NSEI) and Nifty 100 (^CNX100) benchmarks used to calculate active tracking errors, CAPM regression betas, and benchmark alphas.
5. `portfolio_holdings.csv` (Asset & Sector Allocation): Programmatically compiled portfolio weights mapping stock-level holdings and sector allocations across the 10 funds, totaling 170 observations.

3. ETL Pipeline & Data Cleaning

The ETL engine ([`scripts/data_ingestion.py`](file:///c:/Users/dhile/OneDrive/Documents/bluestock_capstone%20project/scripts/data_ingestion.py) and [`scripts/prepare_day4_data.py`](file:///c:/Users/dhile/OneDrive/Documents/bluestock_capstone%20project/scripts/prepare_day4_data.py)) parses raw CSV inputs, applies data normalization rules, and exports processed parquet/CSV files.

Critical Cleaning Rules & Transformations:

- Duplicate Removal: Deduplicated transaction files based on natural transaction IDs.
- Zero-Value Filtering: Filtered out invalid transaction rows where transaction amounts or units were negative or zero.
- Date Alignment: Unified date string formats (e.g. DD-MM-YYYY and ISO formats) to standard ISO dates (YYYY-MM-DD).
- MultiIndex Flattening: Addressed a Yahoo Finance API download limitation where downloading single tickers returns MultiIndexed columns. Programmatically flattened columns using `df.columns = df.columns.get_level_values(0)` to prevent serialization shifts.
- Offline Fallbacks: Integrated a deterministic random walk algorithm that generates realistic index sequences matching Nifty trends if network calls to Yahoo Finance fail.

4. SQL Database Design (Star Schema)

The database schema

([sql/schema.sql](file:///c:/Users/dhile/OneDrive/Documents/bluestock_capstone%20project/sql/schema.sql)) implements a classic dimensional model optimized for analytical execution.

Entity-Relationship Diagram

```
erDiagram
    dim_amc ||--o{ dim_scheme : "manages"
    dim_scheme ||--o{ fact_nav_history : "tracks daily NAV"
    dim_scheme ||--o{ fact_investor_transactions : "processes trades"
    dim_scheme ||--o{ fact_scheme_performance : "reports returns"
    dim_date ||--o{ fact_nav_history : "dates"
    dim_date ||--o{ fact_investor_transactions : "dates"
    dim_date ||--o{ fact_scheme_performance : "dates"
    dim_investor ||--o{ fact_investor_transactions : "executes"

    dim_amc {
        integer amc_id PK
        text amc_code UNIQUE
        text amc_name
        integer founded_year
        text website
    }
    dim_scheme {
        integer scheme_code PK
        text scheme_name
        integer amc_id FK
        text category
        text sub_category
        text benchmark
        text launch_date
    }
    dim_date {
        integer date_id PK
        text full_date UNIQUE
        integer day
        integer month
        text month_name
        integer quarter
        integer year
        integer fiscal_year
    }
    dim_investor {
        text investor_id PK
        text investor_name
        text investor_type
        text city
        text state
        text risk_profile
    }
    fact_nav_history {
        integer nav_id PK
        integer scheme_code FK
        integer date_id FK
        real nav
        real nav_change
        real nav_pct_change
    }
    fact_investor_transactions {
        text txn_id PK
        text investor_id FK
        integer scheme_code FK
        integer date_id FK
        text transaction_type
    }
```

```
real amount_inr
real units
real nav_at_txn
}
fact_scheme_performance {
integer perf_id PK
integer scheme_code FK
integer date_id FK
real returns_1y
real returns_3y
real returns_5y
real sharpe_ratio
real expense_ratio
real aum_cr
}
```

Analytical Views Included:

- `v_latest_nav`: Fetches the most recent Net Asset Value and absolute change for each scheme to drive dashboard indicator cards.
- `v_monthly_transactions`: Summarizes transaction volume, transaction counts, and average investments grouped by type and date.
- `v_scheme_performance_latest`: Compiles the most recent quarterly snapshot of Returns, Sharpe ratios, and AUM values.

5. Exploratory Data Analysis (EDA) Findings

The EDA module ([scripts/generate_eda.py](file:///c:/Users/dhile/OneDrive/Documents/bluestock_capstone%20project/scripts/generate_eda.py)) generated 17 charts highlighting relationships across performance, expense ratios, AUM distribution, and investor profiles:

1. NAV Time Series Analysis (01_nav_time_series.png): Confirms steady upward trends for HDFC Mid-Cap and SBI Small Cap across 2022–2024, compared to cyclical movements in sectoral funds.
2. Daily Returns Distribution (02_daily_returns_dist.png): Seaborn distributions reveal leptokurtic profiles (fat-tails) indicating frequent extreme movements.
3. Scheme Correlation Matrix (03_correlation_matrix.png): Reveals high returns correlation (Pearson correlation > 0.85) between large cap funds, while the Technology sectoral fund shows a low correlation (~0.42) to the rest of the market.
4. Rolling Volatility Dynamics (06_rolling_volatility.png): Rolling 30-day volatility peaks occurred in Q2 2022 and Q3 2024, with Small Cap funds showing the highest overall volatility spikes.
5. Expense vs. Returns scatter (10_expense_vs_returns.png): Demonstrates a negative relationship between high expense ratios and long-term returns, confirming that higher fund management costs can drag down net performance.
6. SIP Inflow Area Chart (12_sip_inflow_trend.png): Identifies structural growth in SIP volumes, with cumulative monthly retail inflows peaking in Q4 2024.

6. Performance Analytics & Ratios

The performance analysis engine compiles annualized risk-adjusted statistics relative to an annual risk-free rate of 6.5% (\$6.5% / 252\$ daily):

Ratios & Formulas Applied:

- Compounded Annual Growth Rate (CAGR):

$$\text{CAGR} = \left(\frac{\text{Ending NAV}}{\text{Beginning NAV}} \right)^{\frac{365.25}{\text{Days}}} - 1$$

- Sharpe Ratio:

$$\text{Sharpe} = \frac{\text{Mean}(R_p - R_f)}{\sigma_p} \times \sqrt{252}$$

- Sortino Ratio:

$$\text{Sortino} = \frac{\text{Mean}(R_p - R_f)}{\sigma_d} \times \sqrt{252}$$

where σ_d is the standard deviation of negative excess returns.

- CAPM Alpha & Beta (Benchmark: Nifty 100):

Daily excess return linear regressions:

$$R_p - R_f = \alpha + \beta (R_b - R_f)$$

Annualized Alpha: $\alpha_{\text{annual}} = \alpha_{\text{daily}} \times 252$.

- Maximum Drawdown (MDD):

$$\text{Drawdown}_t = \frac{\text{Peak}_t - \text{NAV}_t}{\text{Peak}_t}, \quad \text{MDD} = \max(\text{Drawdown}_t)$$

- Tracking Error:

$$\text{Tracking Error} = \sigma(R_p - R_b) \times \sqrt{252}$$

Weighted Scorecard Methodology (Day 4)

Ranks all 10 funds across five factors to assign a final rank:

- 30% Weight: 3-Year Return Rank (Higher CAGR is better)
- 25% Weight: Sharpe Ratio Rank (Higher risk-adjusted return is better)
- 20% Weight: Alpha Rank (Higher active return is better)
- 15% Weight: Expense Ratio Rank (Lower fee is better / Inverse rank)
- 10% Weight: Max Drawdown Rank (Lower loss magnitude is better / Inverse rank)

Performance Scorecard Leaderboard

AMFI Code	Scheme Name	3Y CAGR	Sharpe Ratio	CAPM Beta	Annualized Alpha	Max Drawdown	Tracking Error	Expense Ratio	Score	Final Rank
120594	DSP Top 100 Equity - Dir Growth	39.78%	1.22	-0.04	28.88%	-24.73%	27.56%	0.68%	1.90	#1
120503	Mirae Asset Large Cap - Dir Growth	33.15%	1.07	-0.03	23.87%	-17.47%	26.34%	0.36%	2.05	#2
119551	Axis Bluechip Fund - Dir Growth	29.86%	0.96	0.02	21.23%	-23.78%	26.13%	0.81%	3.65	#3
120716	Canara Robeco Bluechip - Dir Growth	4.72%	0.01	-0.03	0.28%	-24.11%	25.29%	0.77%	5.10	#4
122639	SBI Small Cap Fund - Dir Growth	14.66%	0.41	0.05	9.17%	-31.97%	26.32%	1.18%	5.20	#5
118989	ICICI Pru Technology - Dir Growth	2.13%	-0.09	0.04	-2.22%	-29.88%	25.78%	1.16%	6.45	#6
120465	Nippon India Large Cap - Dir Growth	-2.15%	-0.24	0.06	-5.94%	-27.37%	27.49%	0.76%	6.65	#7
119598	Kotak Bluechip Fund - Dir Growth	-2.60%	-0.28	-0.00	-6.40%	-42.49%	26.95%	0.43%	7.50	#8
135781	Parag Parikh Flexi Cap - Dir Growth	-3.97%	-0.36	-0.09	-7.56%	-35.11%	26.78%	0.34%	7.85	#9
125497	HDFC Mid-Cap Opp - Dir Growth	-5.71%	-0.44	0.08	-10.18%	-49.77%	25.38%	0.34%	8.65	#10

7. Advanced Risk Analytics (Day 6)

Advanced risk analytics provide deeper evaluations of tail risk, portfolio concentration, and investor behavior.

A. Tail Risk Metrics (VaR & CVaR)

Historical Value at Risk (VaR 95%) and Conditional VaR (CVaR 95%) were calculated using daily returns series:

- ICICI Prudential Technology Fund shows elevated tail risk (95% Daily Historical VaR = 2.33% and CVaR = 3.19%), which is expected for concentrated sector-focused portfolios.
- Mirae Asset Large Cap Fund shows a lower tail risk (95% Daily Historical VaR = 2.08% and CVaR = 2.76%), demonstrating the risk-mitigating effect of large-cap diversification.

B. Sector Concentration Analysis (HHI)

Using the Herfindahl-Hirschman Index (HHI) to measure sector-level diversification:

- High Concentration (HHI = 7,338.00): ICICI Prudential Technology Fund has a high HHI due to its 85% concentration in the technology sector.
- Moderate Concentration (HHI = 1,827.16): Standard diversified funds (Large Cap/Flexi Cap) maintain HHIs in this range, diversified across Financial Services, Energy, Tech, Consumer Goods, and Healthcare.
- Low Concentration (HHI = 1,327.42): HDFC Mid-Cap Opportunities Fund shows high sector diversification, holding smaller positions across a wider range of mid-cap companies.

C. SIP Continuity & Cohorts

- SIP Continuity: Active accounts show high continuity rates (average active rate = 87.2%), while inactive accounts show lower continuity rates before churning (average active rate = 94.8% prior to cancellation, with average payment streaks of ~1 month before stopping).
- Cohort Vintages: Q1 2022 and Q3 2023 cohorts represent the largest aggregate investor capital inflows. Net redemption rates are lower among aggressive risk-profile investors compared to conservative investors, who tend to redeem during market volatility.

8. Power BI Dashboard Visual Design

The Power BI Dashboard layout is configured with a high-end dark-mode theme to prioritize data visualization readability.

Page-by-Page Specifications:

Page 1: Industry Overview

- KPI Panel: Four indicators showing Total Industry AUM (■24,850 Cr), SIP Inflow YTD (■4,250 Cr), Active Folios (1.42M), and SIP Contribution % (34.8%).
- AUM Trend: Line and Clustered Column Chart tracking monthly inflows alongside total AUM.
- AMC Concentration: Treemap showing AMC market share.

Page 2: Fund Performance

- Risk-Return Scatter Plot: X-axis = Beta, Y-axis = 3-Year CAGR, Bubble size = AUM (Cr).
- NAV vs. Benchmark: Line chart comparing dynamic NAV performance to the Nifty 100 index (both re-indexed to a base value of ■100).
- Fund Scorecard Table: Interactive scorecard table with underlaid data bars for Sharpe ratio and a color scale for Max Drawdown.

Page 3: Investor Analytics

- State-wise Investments: Indian state shape map showing investment volume using a color scale.
- Transaction Type Split: Donut chart showing transaction distributions (SIP, Lumpsum, Redemption, Switch).
- Age Demographics: Column chart grouping investor folios by age cohort.

Page 4: SIP & Market Trends

- SIP vs. Nifty Trend: Dual-Y Axis line chart comparing monthly SIP inflows with the Nifty 50 close price.
- Inflow Matrix Heatmap: A color-scaled matrix showing net inflows by subcategory and quarter.

9. Key Findings, Recommendations, & Limitations

Strategic Conclusions:

1. Concentration Risk: The Technology Sectoral fund's high concentration ($HHI > 7,000$) results in higher tail risk (larger VaR/CVaR), while diversified large cap funds show better downside protection.
2. Manager Alpha: Active managers in the mid-cap and small-cap segments generated significant Alpha during the 2023–2024 bull runs, compensating for their higher portfolio drawdowns.
3. Investor Behavior: Cohort retention is higher among investors who utilize SIPs, confirming that systematic investment plans encourage longer-term capital retention.

Recommendations for Portfolio Managers:

- Dynamic Risk Budgeting: Adjust sectoral allocations when the HHI rises above 2,500 to mitigate tail risk.
- Promote SIPs: Encourage systematic investment plans to extend investor holding periods and stabilize fund AUM.
- Enhance Capital Preservation: Conservative portfolios should focus on Large Cap allocations to maintain drawdowns above -20%.

Platform Limitations:

- Historical Horizon: The daily NAV data is limited to 3 years (2022–2024). Extending the historical range to 5–10 years would improve the accuracy of the risk models.
- Transaction Granularity: The transaction records are simplified; incorporating transaction costs and tax implications (e.g. exit loads) would improve tracking accuracy.

10. Conclusion

The Mutual Fund Analytics & Quantitative Risk Platform provides an integrated solution for data-driven portfolio management. By combining structured data modeling, automated ETL processes, advanced quantitative risk metrics, and interactive dashboard layouts, the platform enables analysts to make better-informed, risk-managed investment decisions.